Chemistry: CYL1010

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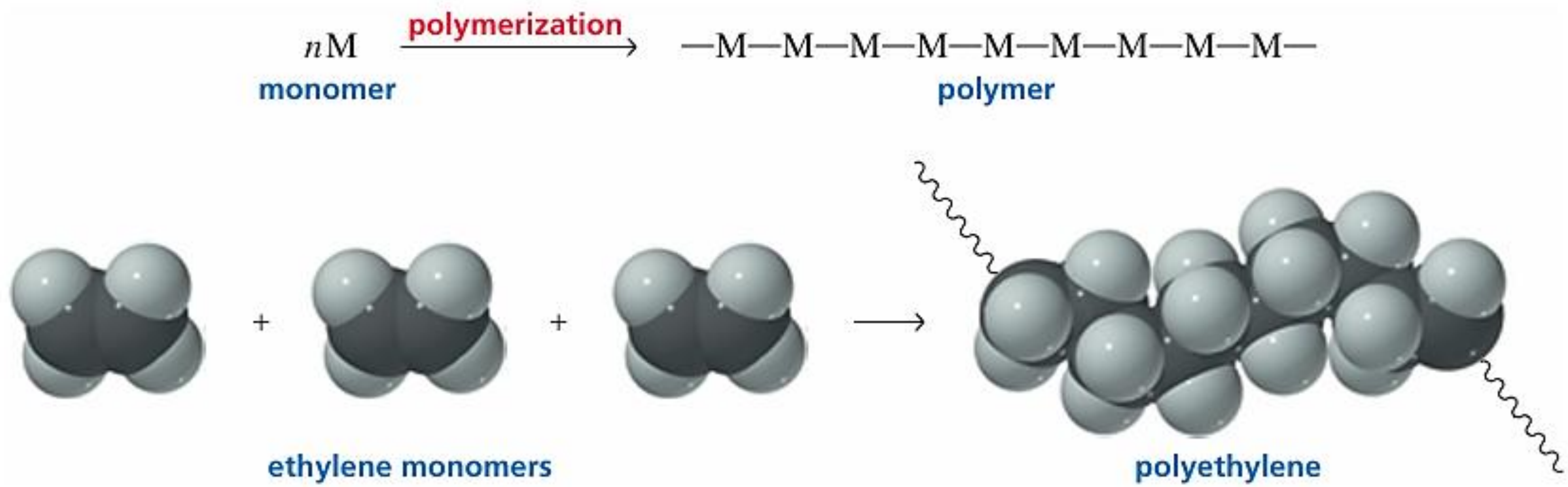
Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

Polymers

A polymer is a large molecule made by linking together repeating units of small molecules called monomers. The process of linking them together is called polymerization.



These giant molecules are interesting because of their physical properties, which make them useful in everyday life such as Compact discs, Rugs, Food wrap, Artificial joints, Super glue, toys, Bottles, Weather stripping, Automobile body parts, Shoe soles, etc.

Polymers can be divided into two broad groups: synthetic polymers and biopolymers (natural polymers).

Synthetic polymers are synthesized by scientists, whereas biopolymers are synthesized by organisms.

Examples of biopolymers are DNA, the storage molecule for genetic information; RNA and proteins, the molecules that induce biochemical transformations; and polysaccharides.

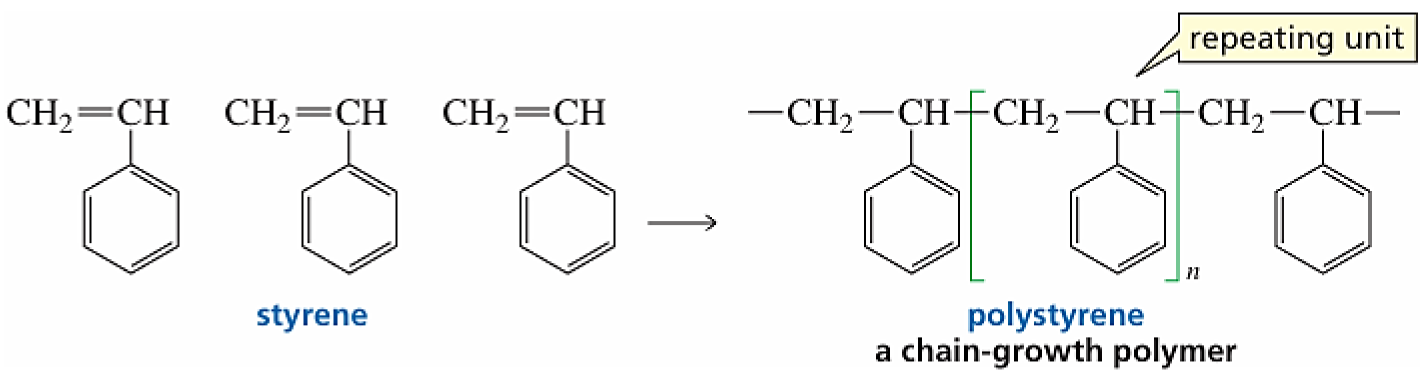
Humans first relied on natural polymers for clothing, wrapping themselves in animal skins and furs. Later, they learned to spin natural fibers into thread and to weave the thread into cloth. Today, much of our clothing is made of synthetic polymers (e.g., nylon, polyester, polyacrylonitrile). Many people prefer clothing made of natural polymers (e.g., cotton, wool, silk), The first plastic—a polymer capable of being molded—was celluloid. Invented in 1856 by Alexander Parke, it was a mixture of nitrocellulose and camphor. Celluloid was used in the manufacture of billiard balls and piano keys, replacing scarce ivory.

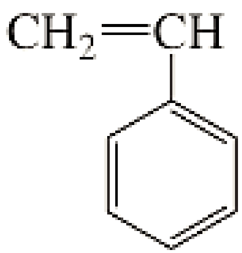
The first synthetic fiber was rayon. In 1865, the French silk industry was threatened by an epidemic that killed many silkworms, highlighting the need for an artificial silk substitute. Louis Chardonnet accidentally discovered the starting material for a synthetic fiber when, while wiping up some spilled nitrocellulose from a table, he noticed long silklike strands adhering to both the cloth and the table. "Chardonnet silk" was introduced at the Paris Exposition in 1891. It was called rayon because it was so shiny that it appeared to give off rays of light.

The first synthetic rubber was synthesized by German chemists in 1917. Their efforts were in response to a severe shortage of raw materials as a result of blockading during World War I.

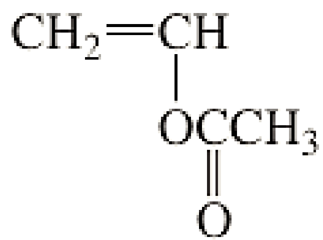
Synthetic polymers can be divided into two major classes, depending on their method of preparation. Chain-growth polymers, also known as addition polymers, are made by chain reactions-the addition of monomers to the end of a growing chain. The end of the chain is reactive because it is a radical, a cation, or an anion.

Polystyrene—used for disposable food containers, insulation, and toothbrush handles, among other things.

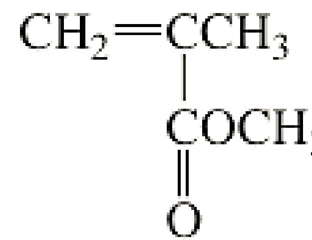




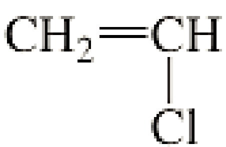
styrene



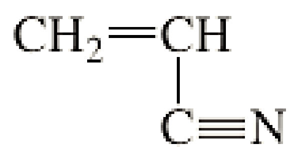
vinyl acetate



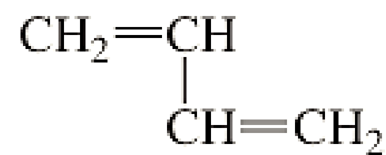
methyl methacrylate



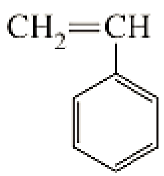
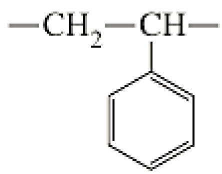
vinyl chloride



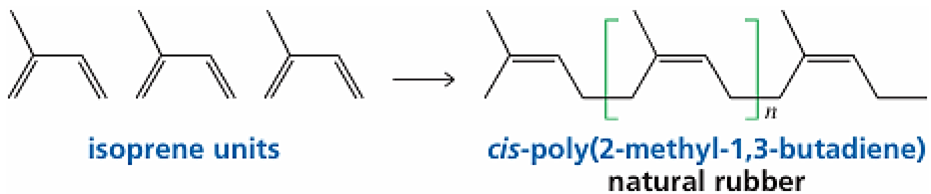
acrylonitrile

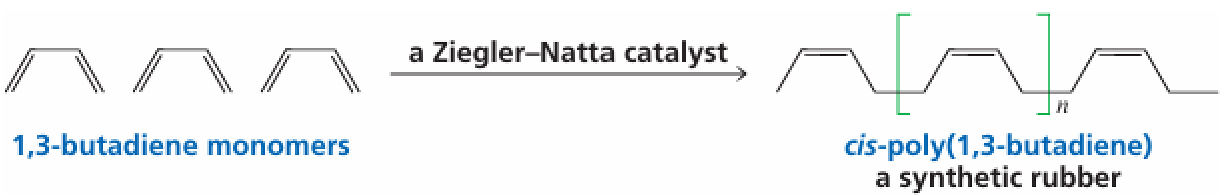


1,3-butadiene

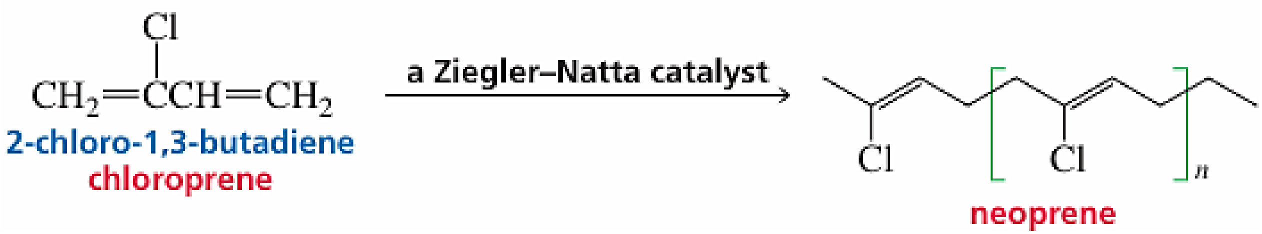
Monomer	Repeating unit	Polymer name	Uses
$CH_2{=}CH_2$	$-CH_2-CH_2-$	polyethylene	film, toys, bottles, plastic bags
$CH_2{=}CH(Cl)$	$-CH_2-CH(Cl)-$	poly(vinyl chloride)	"squeeze" bottles, pipe, siding, flooring
$CH_2{=}CH-CH_3$	$-CH_2-CH(CH_3)-$	polypropylene	molded caps, margarine tubs, indoor/outdoor carpeting, upholstery
		polystyrene	packaging, toys, clear cups, egg cartons, hot drink cups
$CF_2{=}CF_2$	$-CF_2-CF_2-$	poly(tetrafluoroethylene) Teflon®	nonsticking surfaces, liners, cable insulation
$CH_2{=}CH(C{\equiv}N)$	$-CH_2-CH(C{\equiv}N)-$	poly(acrylonitrile) Orlon®, Acrilan®	rugs, blankets, yarn, apparel, simulated fur
$CH_2{=}C(CH_3)-C({=}O)OCH_3$	$-CH_2-C(CH_3)(C({=}O)OCH_3)-$	poly(methyl methacrylate) Plexiglas®, Lucite®	lighting fixtures, signs, solar panels, skylights
$CH_2{=}CH-OC({=}O)CH_3$	$-CH_2-CH(OC({=}O)CH_3)-$	poly(vinyl acetate)	latex paints, adhesives

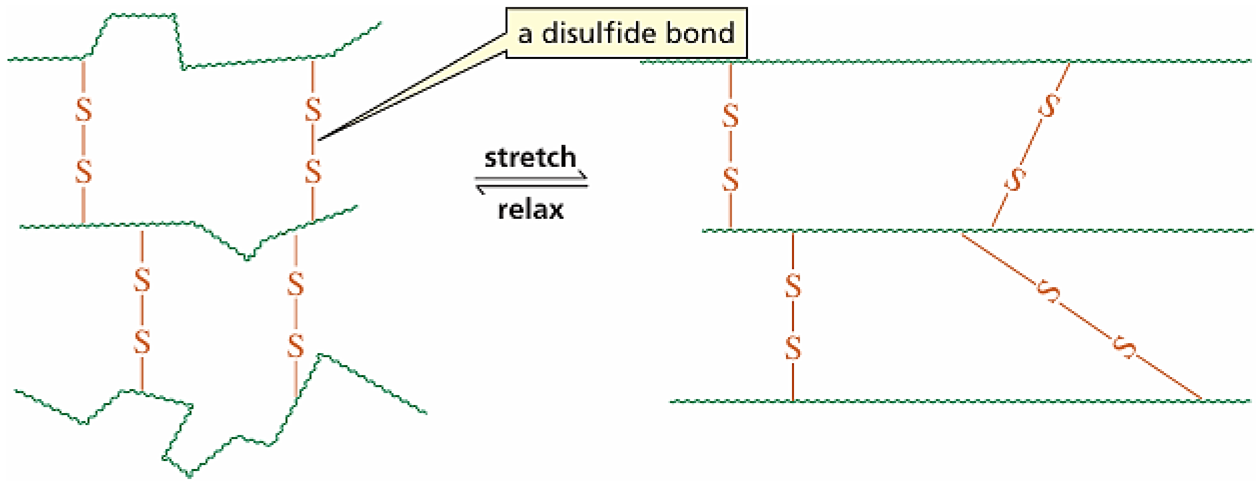
Natural rubber is a polymer of 2-methyl-1,3-butadiene (isoprene). On average, a molecule of rubber contains 5000 isoprene units. All the double bonds in natural rubber are cis. Rubber is a waterproof material because it consists of a tangle of hydrocarbon chains that have no affinity for water. Charles Macintosh, a Scotsman, was the first to use rubber as a waterproof coating for raincoats





Neoprene is a synthetic rubber made by polymerizing 2-chloro-1,3-butadiene in the presence of a Ziegler–Natta catalyst that causes all the double bonds in the polymer to have the trans configuration. Neoprene is used to make wet suits, shoe soles, tires, hoses, and coated fabrics.



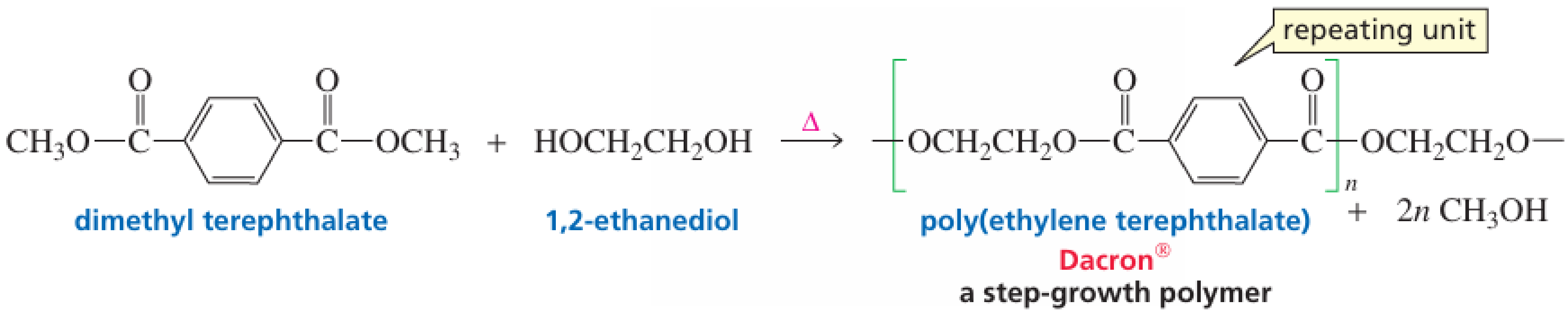


The rigidity of rubber is increased by cross-linking the polymer chains with disulfide bonds. When rubber is stretched, the randomly coiled chains straighten out and orient themselves along the direction of the stretch

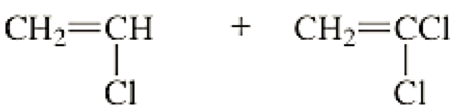
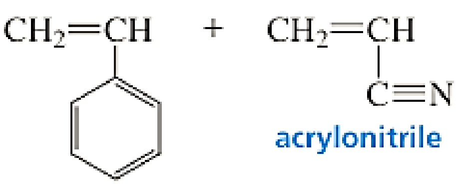
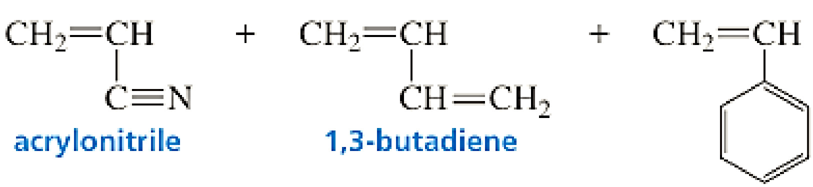
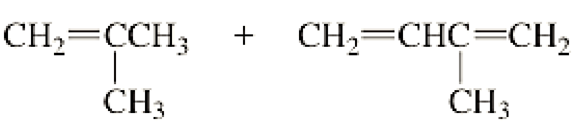
The physical properties of rubber can be controlled by regulating the amount of sulfur used in the vulcanization process. Rubber made with 1–3% sulfur is soft and stretchy and is used to make rubber bands. Rubber made with 3–10% sulfur is more rigid and is used in the manufacture of tires.

Step-growth polymers, also called condensation polymers, are made by combining two molecules while, in most cases, removing a small molecule, generally water or an alcohol. The reacting molecules have reactive functional groups at each end.

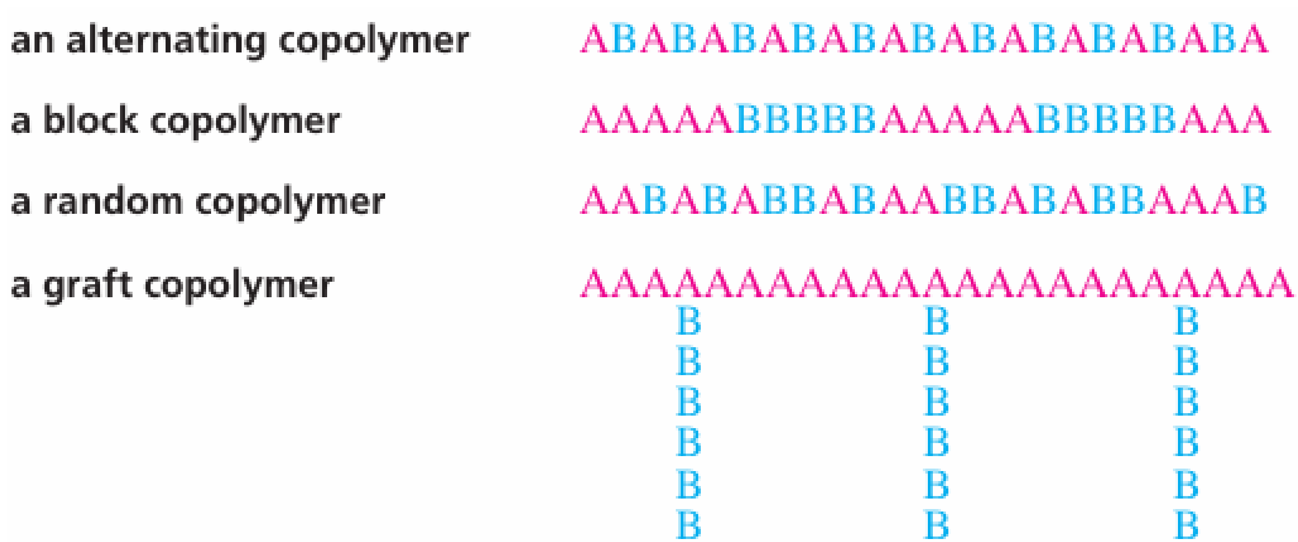
Dacron® is an example of a step-growth polymer.



The polymers we have discussed so far are formed from only one type of monomer and are called homopolymers. Often, two or more different monomers are used to form a polymer. The resulting product is called a copolymer. Both chain-growth polymers and step-growth polymers can be copolymers. Many of the synthetic polymers used today are copolymers.

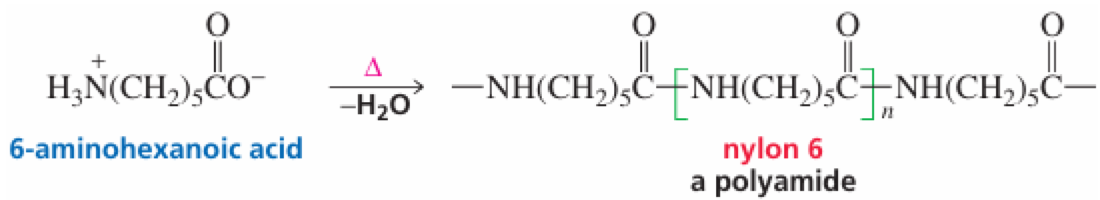
Monomer	Copolymer name	Uses
 vinyl chloride + vinylidene chloride	Saran	film for wrapping food
 styrene + acrylonitrile	SAN	dishwasher-safe objects, vaccum cleaner parts
 acrylonitrile + 1,3-butadiene + styrene	ABS	bumpers crash helmets, telephones, luggage
 isobutylene + isoprene	butyl rubber	inner tubes, balls, inflatable sporting goods

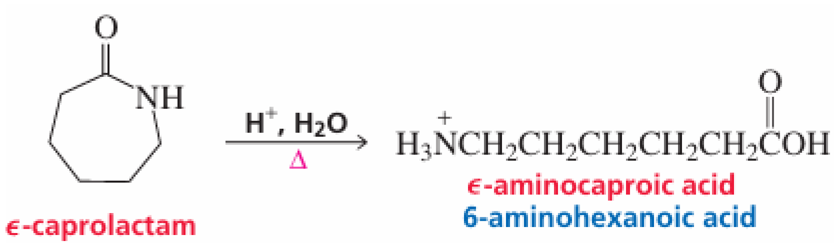
There are four types of copolymers. In an alternating copolymer, the two monomers alternate. In a block copolymer, there are blocks of each kind of monomer. In a random copolymer, the distribution of monomers is random. A graft copolymer contains branches derived from one monomer grafted onto a backbone derived from another monomer.

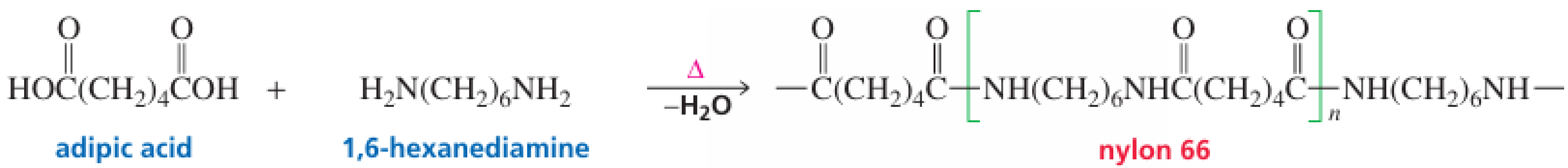


Step Growth Polymers

Polyamides

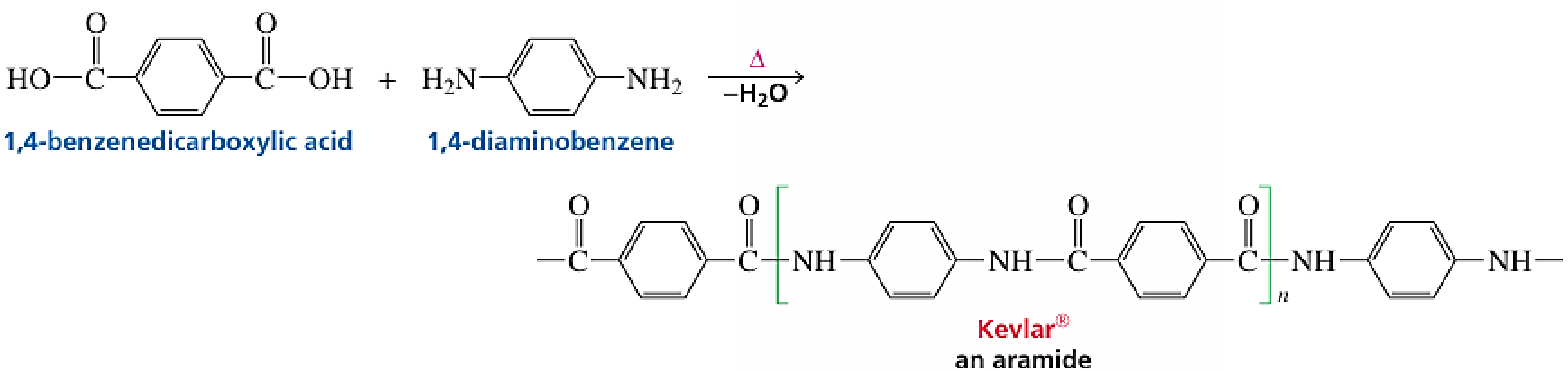


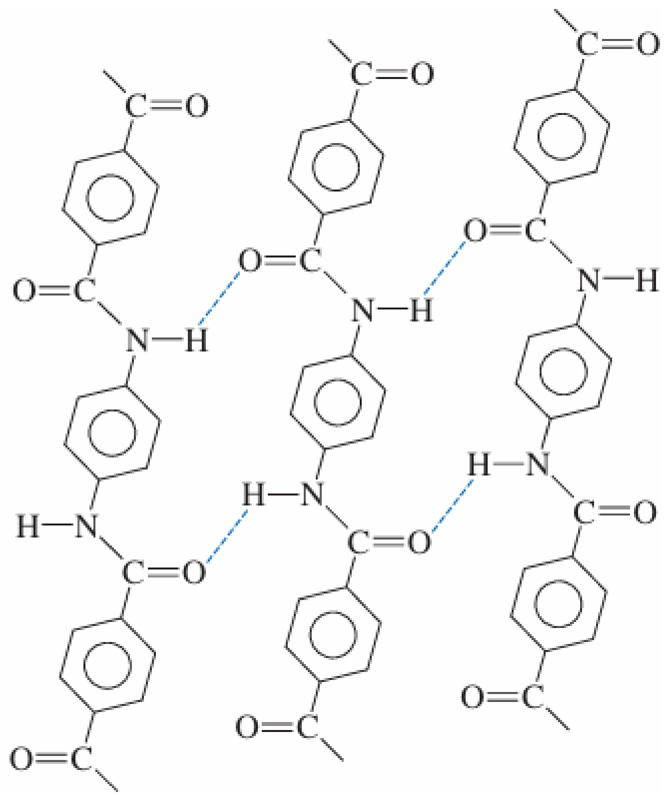




Nylon first found wide use in textiles and carpets. Because it is resistant to stress, it is now used in many other applications, such as mountaineering ropes, tire cords and fishing lines, and as a substitute for metal in bearings and gears. The extended appli cations of nylon precipitated a search for new “super fibers” with super strength and super heat resistance.

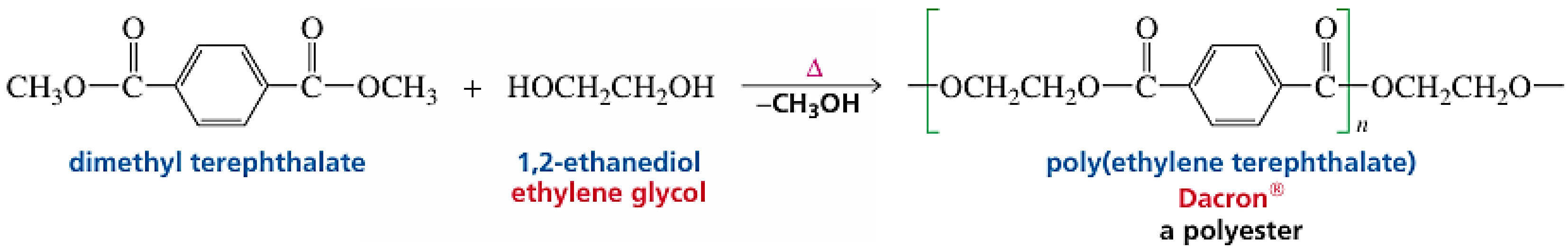
One super fiber is Kevlar®, a polymer of 1,4-benzenedicarboxylic acid and 1,4-diaminobenzene. The incorporation of aromatic rings into polymers has been found to result in polymers with great physical strength. Aromatic polyamides are called aramides. Kevlar® is an aramide with a tensile strength greater than that of steel. Army helmets are made of Kevlar®, which is also used for lightweight bullet proof vests and high-performance skis. Because it is stable at very high temperatures, it is used in the protective clothing worn by firefighters



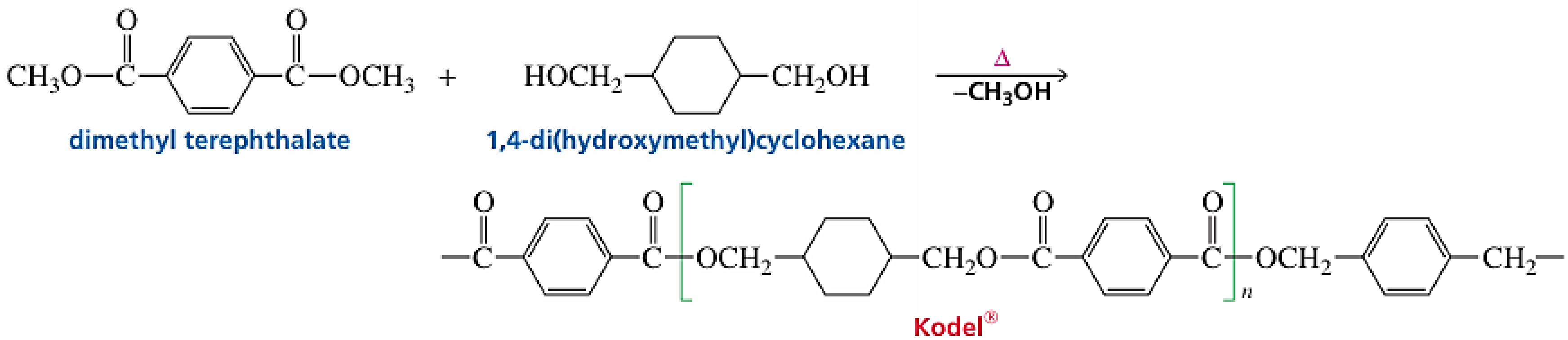


Polyesters

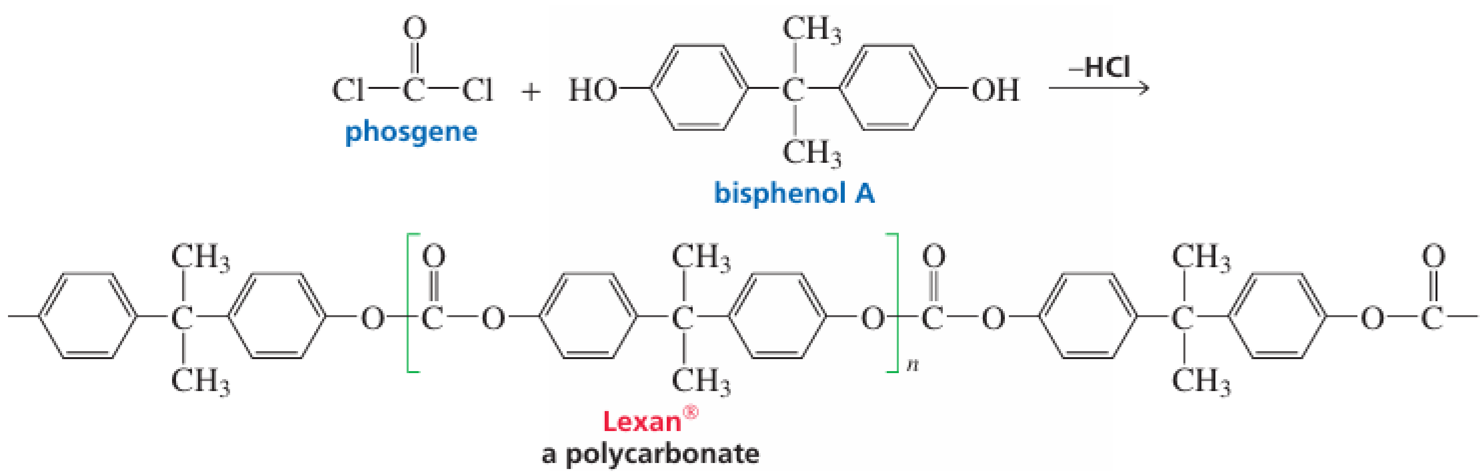
Polyesters are step-growth polymers in which the monomer units are joined together by ester groups. They have found wide commercial use as fibers, plastics, and coatings. The most common polyester is known by the trade name Dacron® and is made by the transesterification of dimethyl terephthalate with ethylene glycol. High resilience, durability, and moisture resistance are the properties of this polymer that contribute to its "wash-and-wear" characteristics.



Kodel® polyester is formed by the transesterification of dimethyl terephthalate with 1,4-di(hydroxymethyl)cyclohexane. The stiff polyester chain causes the fiber to have a harsh feel that can be softened by blending it with wool or cotton.

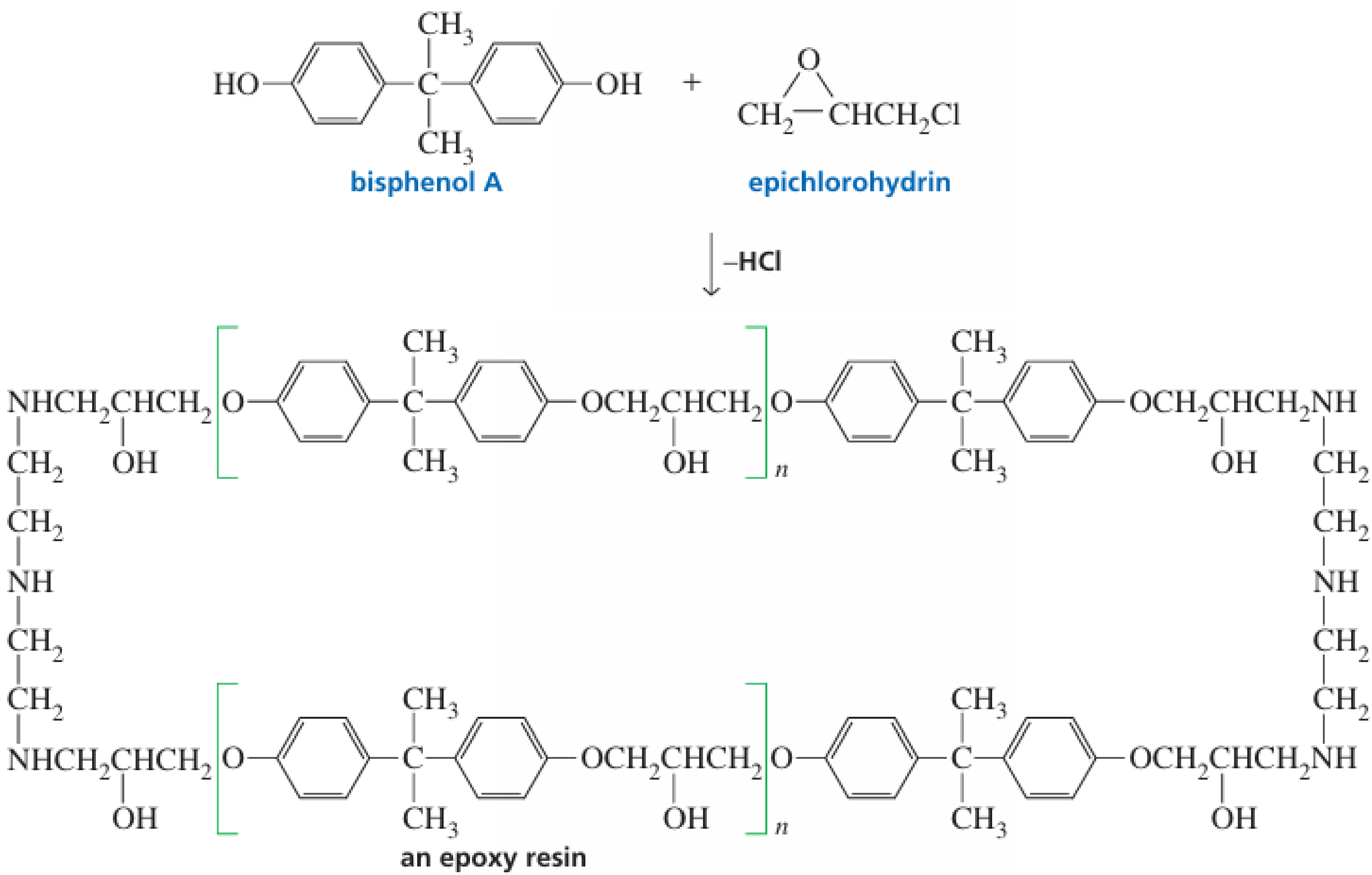


Polyesters with two ester groups bonded to the same carbon are known as polycarbonates. Lexan® is produced by the reaction of phosgene with bisphenol A. Lexan® is a strong, transparent polymer used for bulletproof windows and traffic-light lenses. In recent years, polycarbonates have become important polymers in the automobile industry as well as in the manufacture of compact discs.

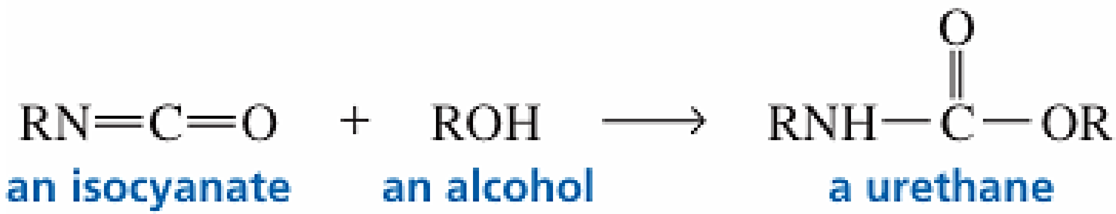


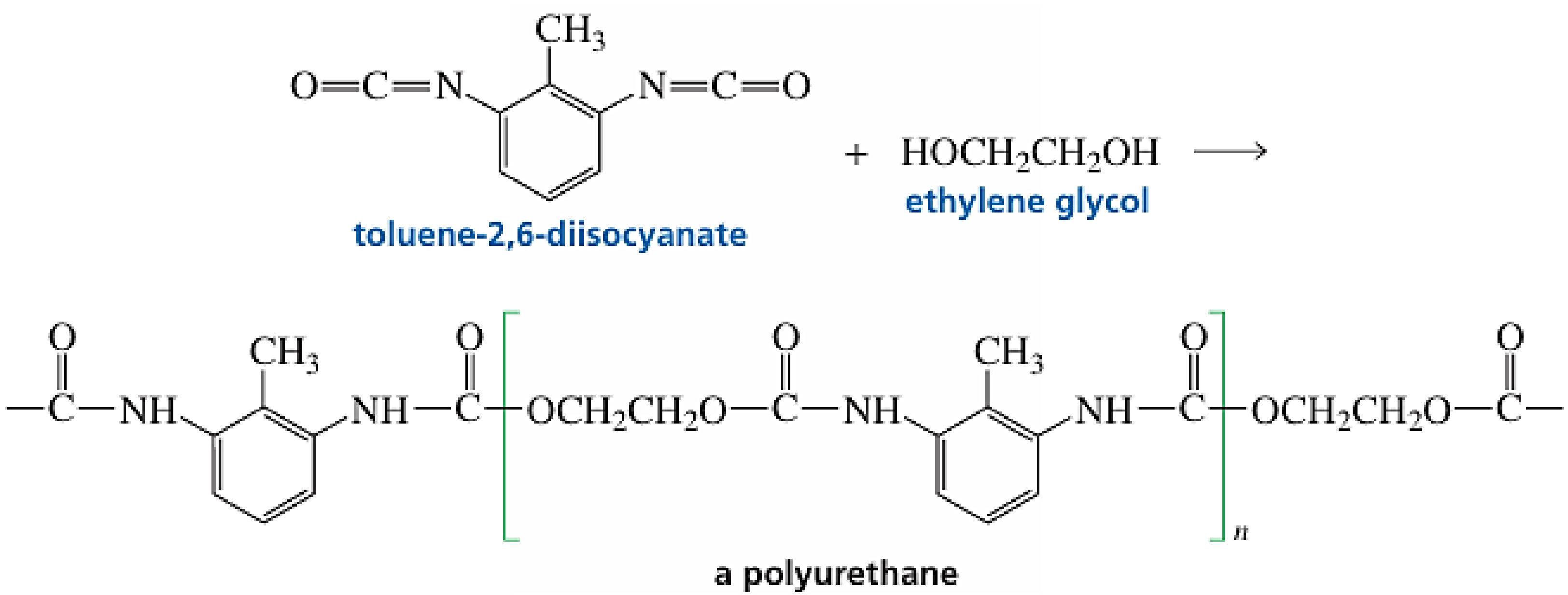
Epoxy Resins

Epoxy resins are the strongest adhesives known. They can adhere to almost any kind of surface and are resistant to solvents and to extremes of temperature. When an epoxy cement is used, a low-molecular-weight prepolymer(the most common is a polymer of bisphenol A and epichlorohydrin) is mixed with a hardener—a compound that will react with the prepolymer to form a cross-linked polymer



Polyurethanes





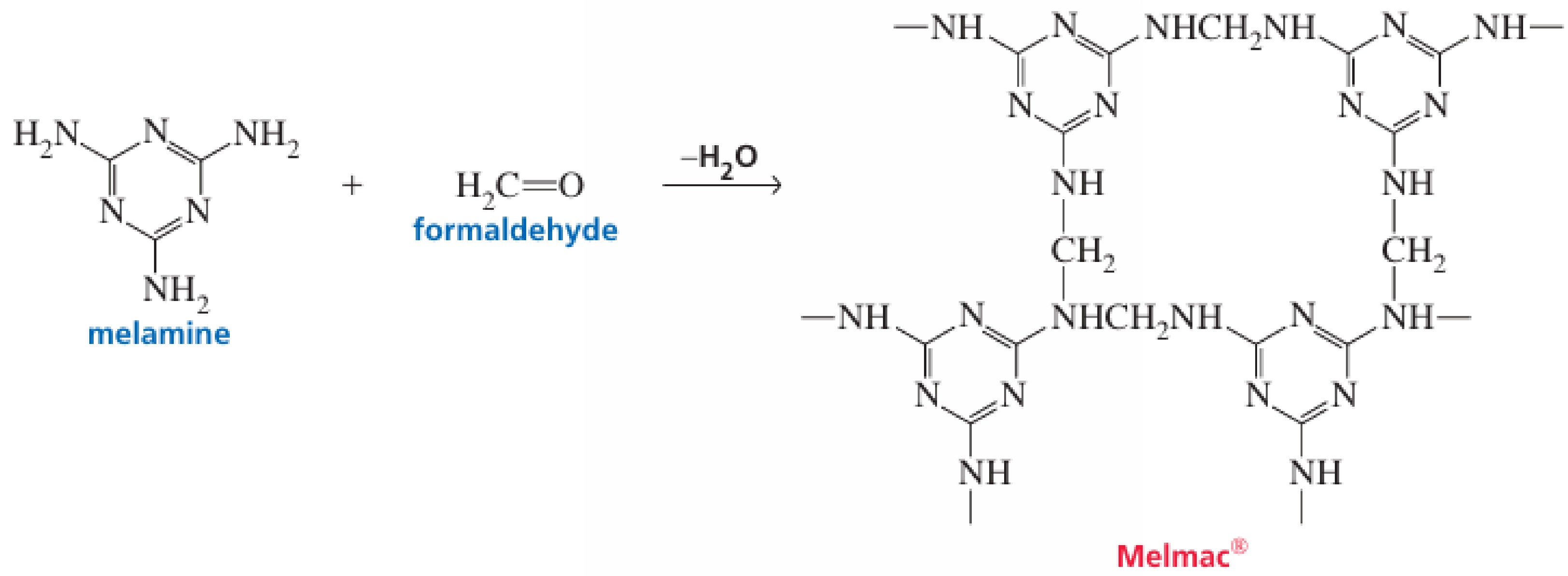
Thermoplastic Polymers

Plastics can be classified according to the physical properties imparted to them by the way in which their individual chains are arranged. Thermoplastic polymers have both ordered crystalline regions and amorphous non-crystalline regions. Thermoplastic polymers are hard at room temperature, but soft enough to be molded when heated, because the individual chains can slip past one another at elevated temperatures.

Thermosetting Polymers

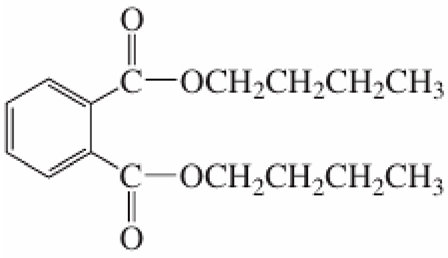
Very strong and rigid materials can be obtained if polymer chains are cross-linked. The greater the degree of cross-linking, the more rigid is the polymer. Such cross-linked polymers are called thermosetting polymers. After they are hardened, they cannot be remelted by heating, because the cross-links are covalent bonds, not intermolecular van der Waals forces.

Melmac®, a highly cross-linked thermosetting polymer of melamine and formaldehyde, is a hard, moisture-resistant material. Because it is colorless, Melmac® can be made into materials with pastel colors. It is used to make lightweight dishes and counter surfaces.



Plasticizers

A plasticizer can be added to a polymer to make it more flexible. A plasticizer is an organic compound that dissolves in the polymer, lowering the attractions between the polymer chains, which allows them to slide past one another. Dibutyl phthalate is a commonly used plasticizer. It is added to poly(vinyl chloride)—normally a brittle polymer— to make products such as vinyl raincoats, shower curtains, and garden hoses



dibutyl phthalate
a plasticizer

Polymer	Monomer	Uses of Polymer
Rubber	Isoprene (1, 2-methyl 1 – 1, 3-butadiene)	Making tyres, elastic materials
BUNA – S	(a) 1, 3-butadiene (b) Styrene	Synthetic rubber
BUNA – N	(a) 1, 3-butadiene (b) Vinyl Cyanide	Synthetic rubber
Teflon	Tetra Fluoro Ethane	Non-stick cookware – plastics
Terylene	(a) Ethylene glycol (b) Terephthalic acid	Fabric
Glyptal	(a) Ethylene glycol (b) Phthalic acid	Fabric
Bakelite	(a) Phenol (b) Formaldehyde	Plastic switches, Mugs, buckets
PVC	Vinyl Cyanide	Tubes, Pipes
Melamine Formaldehyde Resin	(a) Melamine (b) Formaldehyde	Ceramic, plastic material
Nylon-6	Caprolactam	Fabric

Biopolymers

Biopolymers are high-molecular-weight compounds produced by living organisms or synthesized to mimic nature's polymers. They differ from conventional synthetic polymers due to their biodegradability and renewable origins.

Understanding these unique materials is essential due to their central role in biological systems and innovative industrial uses.

Types of Biopolymers

- **Polysaccharides** – Complex carbohydrates like cellulose and starch.
- **Proteins** – Polymers of amino acids, including enzymes and collagen.
- **Nucleic Acids** – DNA and RNA, responsible for genetic information.
- **Synthetic biopolymers** – Examples include polylactic acid (PLA), designed to be eco-friendly alternatives to plastics.

Characteristics of Biopolymers

- Composed of repeat units (monomers) linked by covalent bonds.
- Naturally biodegradable, causing less environmental harm than traditional plastics.
- Can possess high mechanical strength, elasticity, or specific biological functions.
- Used in food packaging, medical devices, and tissue engineering applications.

Common Biopolymers Examples

- **Cellulose** – Most abundant organic polymer, forms plant cell walls.
- **Chitosan** – Extracted from shellfish; used in water purification and wound healing.
- **Gelatin** – Obtained from collagen, widely used in pharmaceuticals and foods.
- **Polylactic acid** (PLA) – Biodegradable plastic substitute, prominent in single-use products.

Some advanced applications also come with challenges, like unwanted deposits or reactions in living systems, requiring innovative biopolymers removal or biopolymers removal surgery, particularly in cosmetic or medical contexts. Specialized biopolymers facility and professionals handle these procedures to ensure safety.

Properties and Uses of Biopolymers

Biopolymers showcase versatile physical and chemical properties that make them valuable for different fields. Their composition determines their function and potential for customization.

Key Properties

•High degree of polymerization—long chains or networks.

•Thermal sensitivity—many degrade at high temperatures.

•Excellent biocompatibility for medical implants or scaffolds.

Biopolymers Uses

•Packaging, especially food-safe and biodegradable films.

•Biomedical devices: sutures, scaffolds, wound dressings.

•Agricultural films and slow-release fertilizers.

•Drug delivery and tissue engineering research.

Degradation, Removal, and Safety

Biopolymers naturally degrade into simpler substances under environmental or biological conditions. However, some synthetic versions or improperly placed biopolymers (e.g., in cosmetic procedures like biopolymers buttocks) may require removal. Techniques include:

•Enzymatic or hydrolytic breakdown.

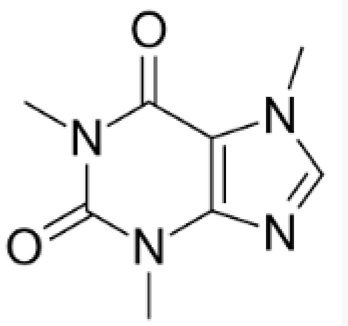
•Surgical removal, often in specialized settings (biopolymers removal surgery).

•Safe disposal in a certified biopolymers facility.

Common Everyday Drugs

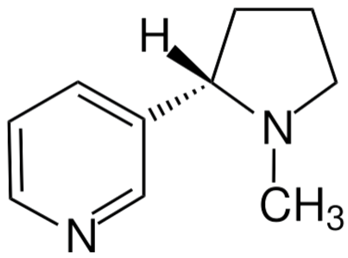
Stimulants

Caffeine



A natural stimulant found in coffee/tea; increases alertness by blocking adenosine receptors.

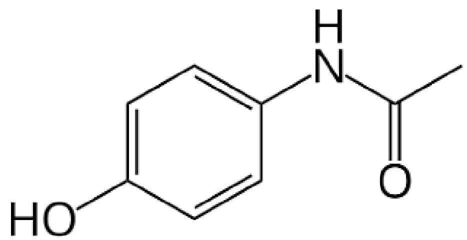
Nicotine



Psychoactive compound in tobacco; activates acetylcholine receptors.

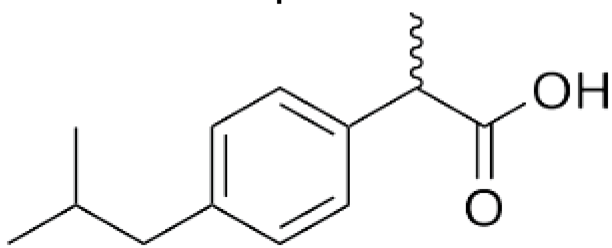
Analgesics / Pain Relievers

Acetaminophen (Paracetamol)



Non-opioid analgesic and fever reducer; gentle on the stomach.

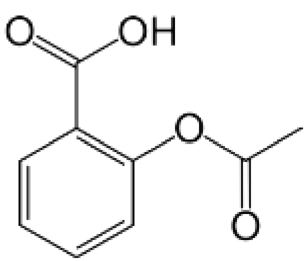
Ibuprofen



(*R*/*S*)-ibuprofen

NSAID that reduces inflammation, swelling, fever, and pain.

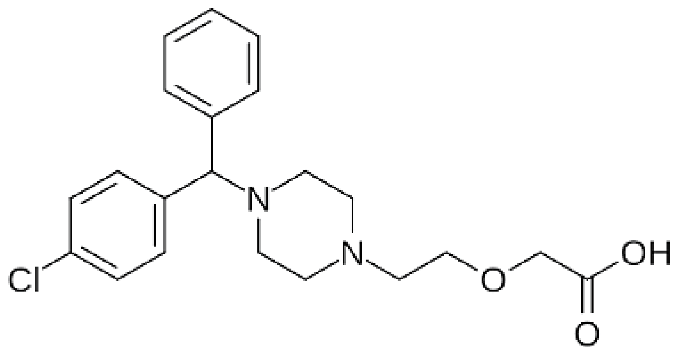
Aspirin (Acetylsalicylic Acid)



Classic NSAID; prevents blood clotting at low doses.

Antihistamines

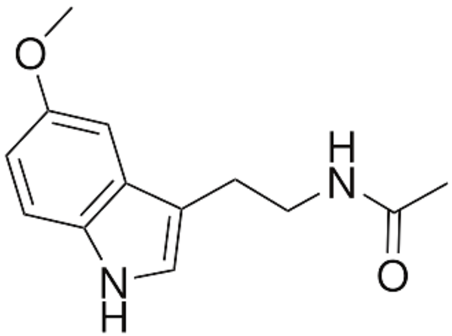
Cetirizine



Second-generation antihistamine; non-drowsy allergy relief.

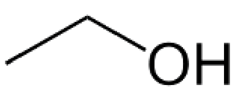
CNS / Hormonal

Melatonin



Naturally occurring sleep-regulating hormone; also sold as a supplement.

Ethanol (Alcohol)

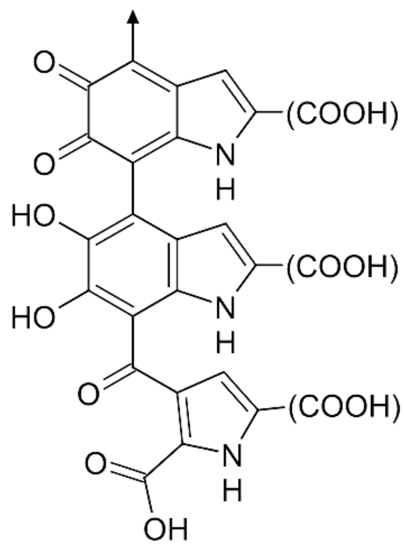


Mild CNS depressant found in beverages; affects GABA and dopamine systems.

Photoactive Molecules in Everyday Life

Natural Pigments

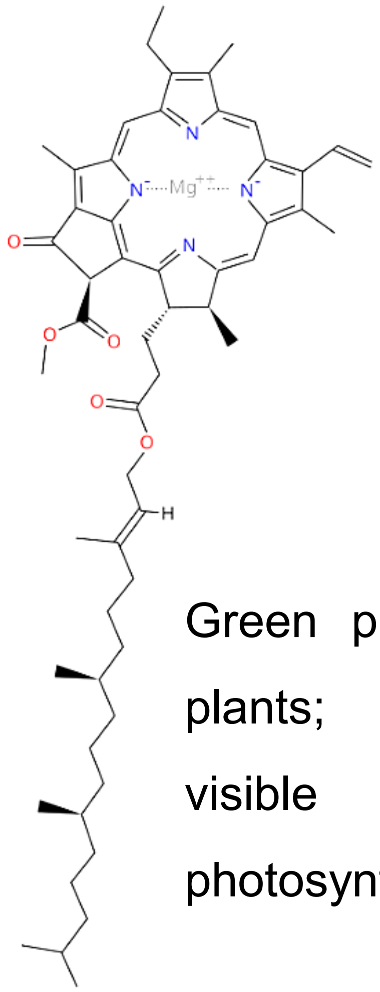
Melanin



One possible structure of eumelanin

Skin/hair pigment that absorbs UV light and protects cells.

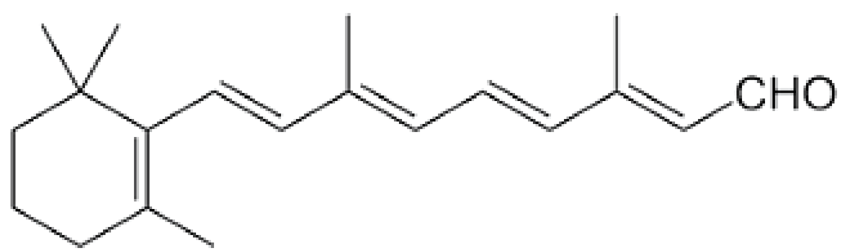
Chlorophyll a



Green pigment in plants; absorbs visible light for photosynthesis.

Vision-Related Photoactive Molecules

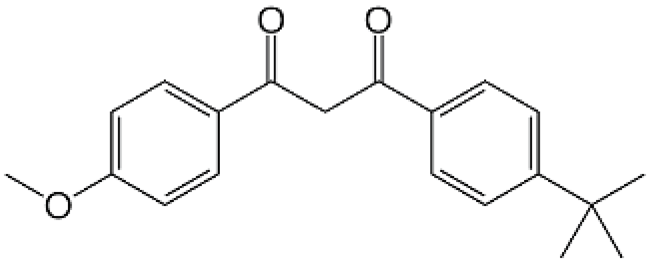
Retinal (11-cis-retinal)



Light-sensitive chromophore in rhodopsin; triggers vision by isomerizing.

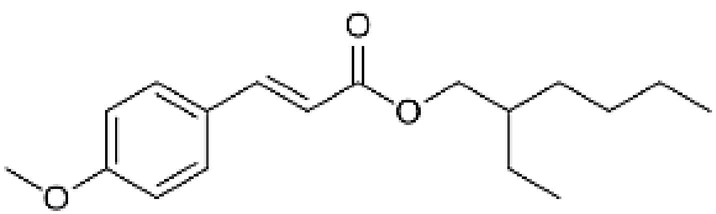
Sunscreen UV-absorbing molecules

Avobenzone



Organic molecule that absorbs UVA radiation (320–400 nm).

Octinoxate



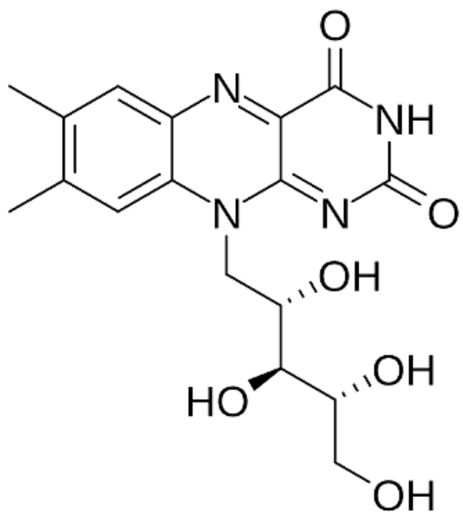
UV-B absorbing sunscreen agent.

Titanium dioxide (TiO_2)

Physical UV blocker used in sunscreens and self-cleaning surfaces.

Food and Vitamin Photoactives

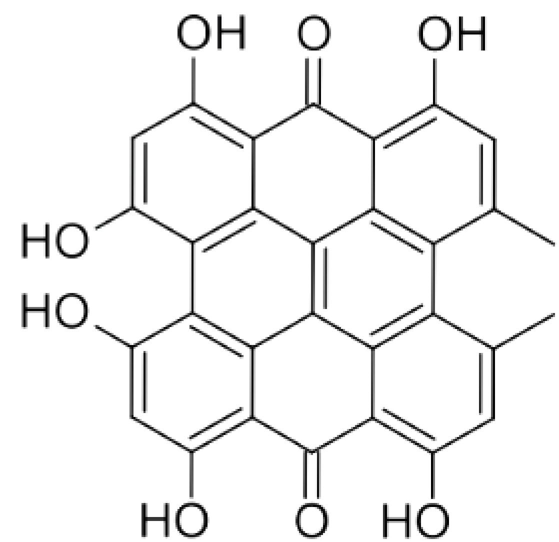
Riboflavin (Vitamin B_2)



Light-sensitive vitamin; degrades under sunlight (hence milk in opaque containers).

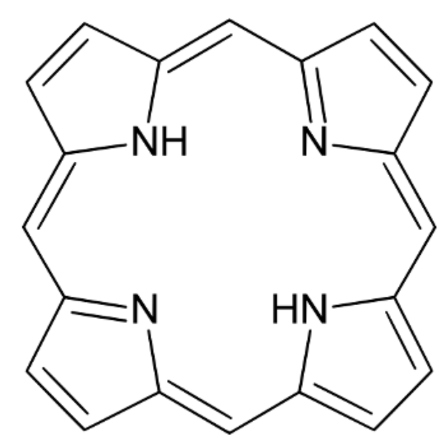
Bioactive and Photoactive

Hypericin



Hypericin is a naturally occurring substance found in the common St. John's Wort (Hypericum species) and can also be synthesized from the anthraquinone derivative emodin. It has traditionally been used throughout the history of folk medicine.

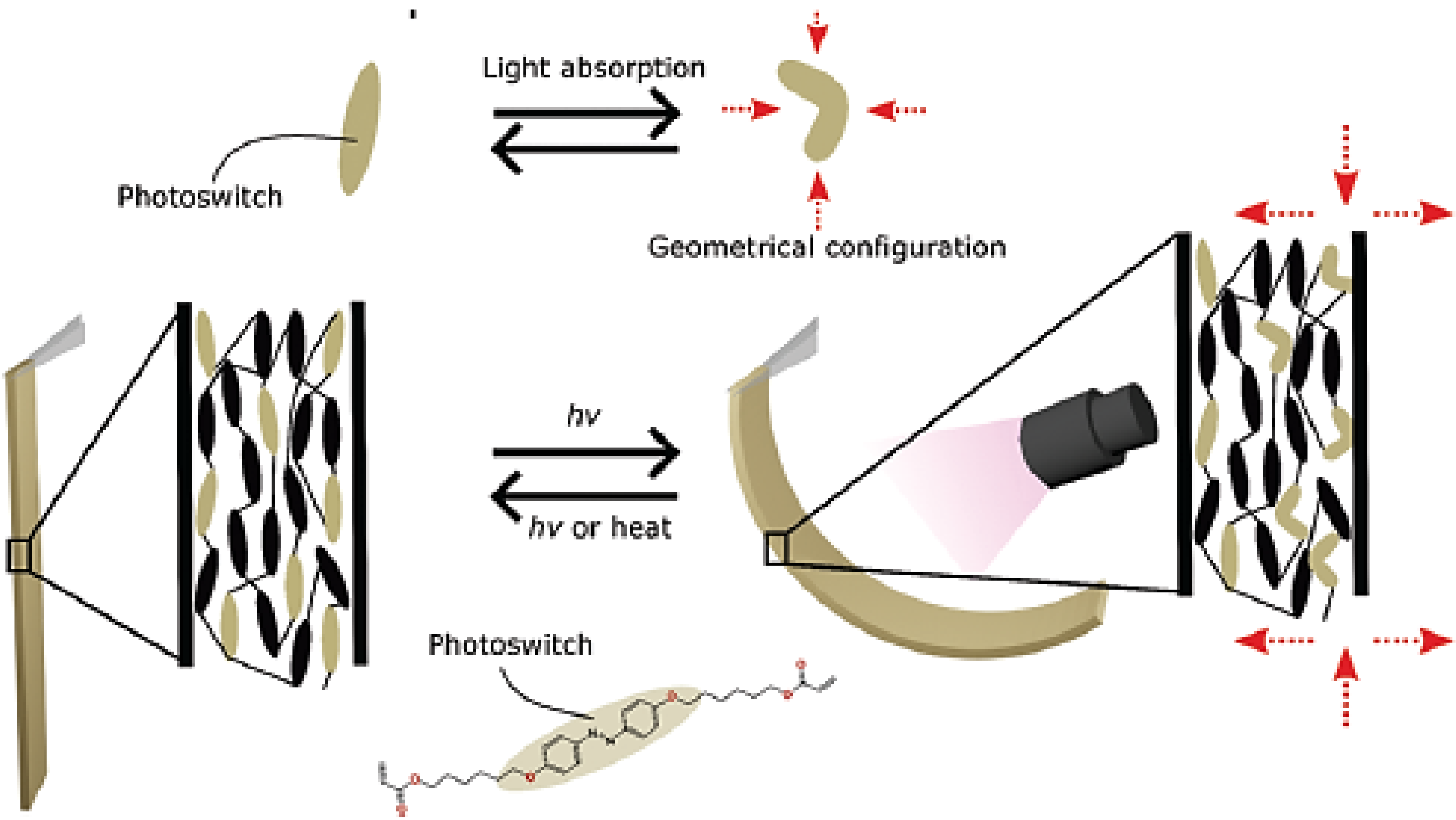
Porphyrins



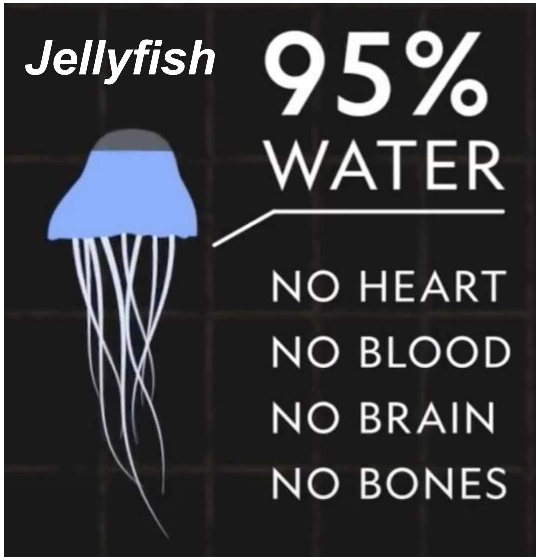
Light-absorbing molecules in heme/chlorophyll; used in photodynamic cancer therapy.

Photoactive molecules in advanced technologies

Photomechanical Switch



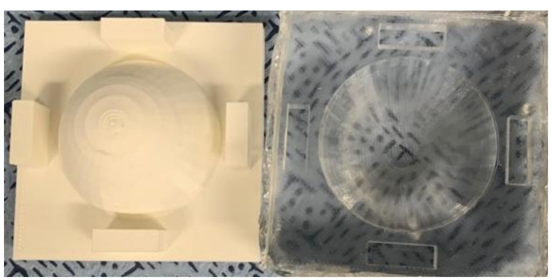
Artificial Life Inspired Robotics

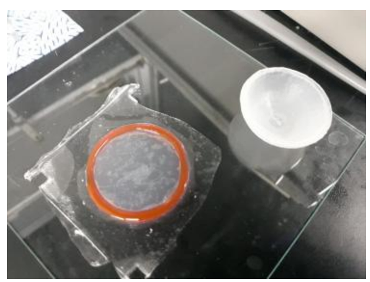


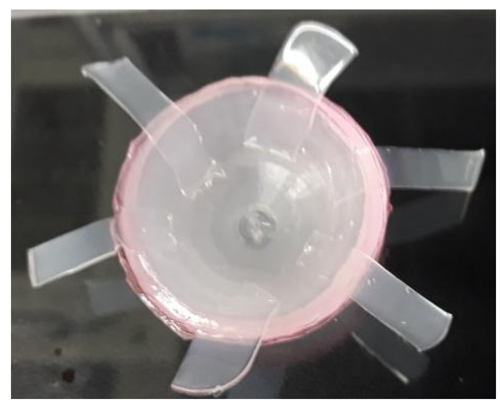
Hydrogel

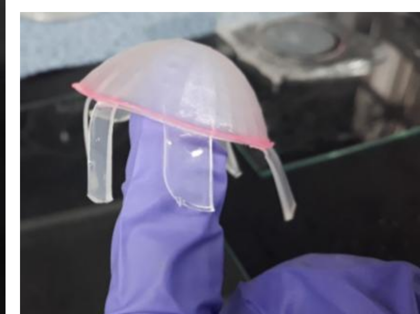
High
water contents

No Heart
No Blood
No Brain
No Bones

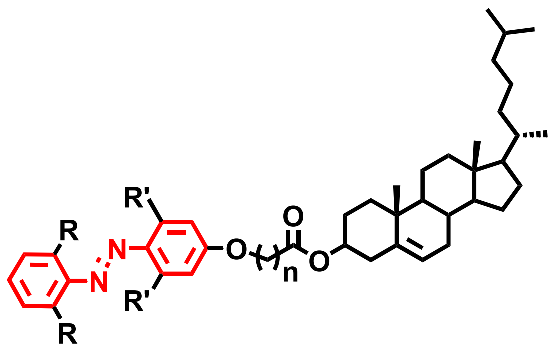


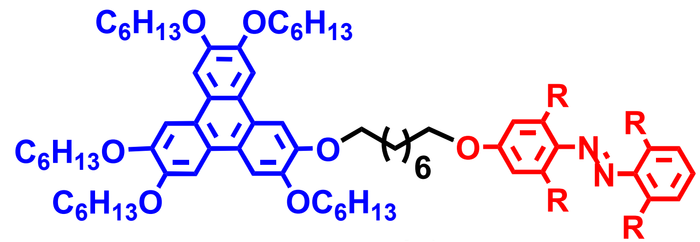


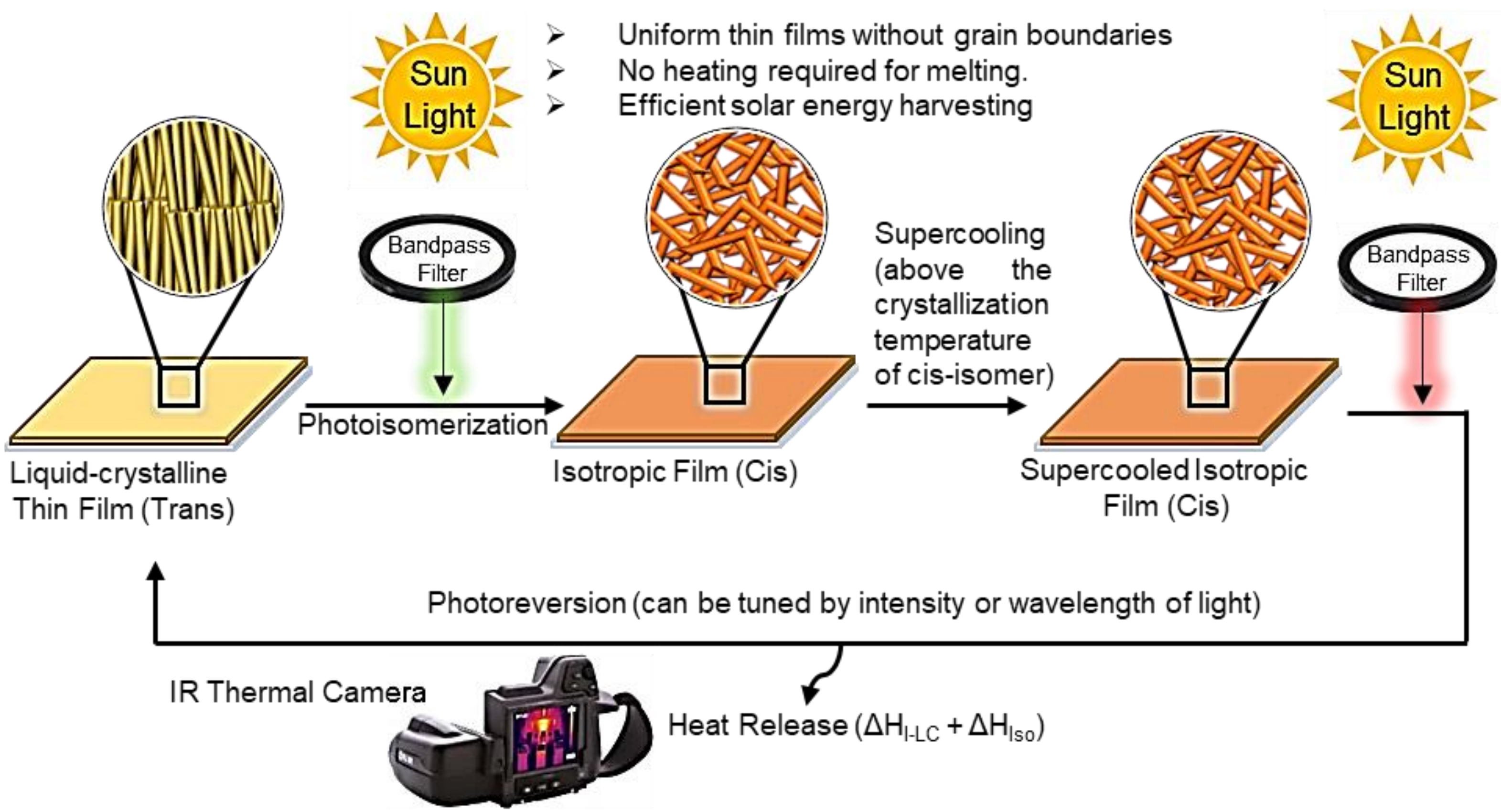




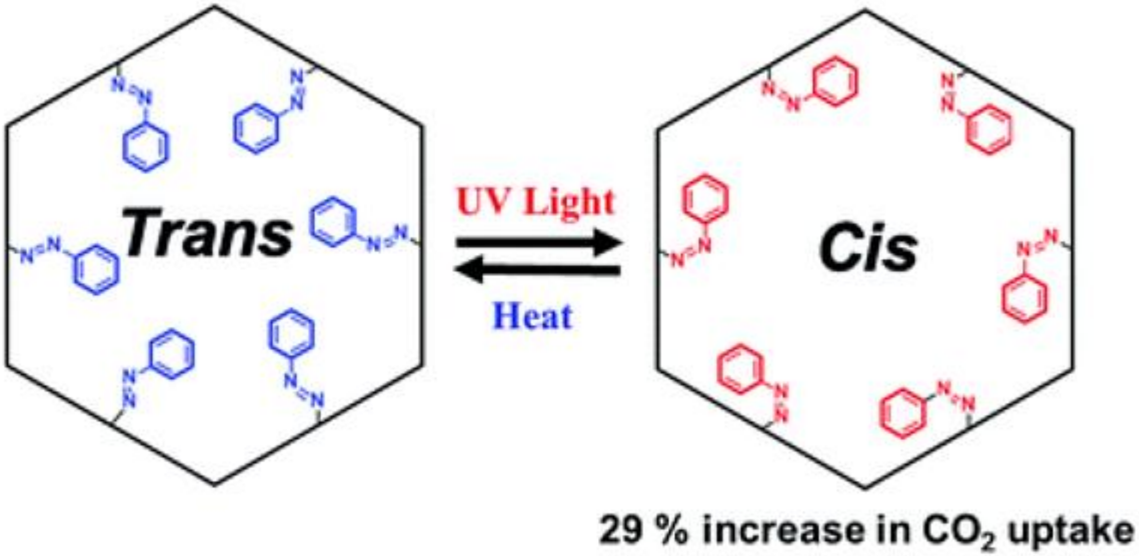
Solar Thermal Fuels







Reversible tuning of pore size and CO_2 adsorption in azobenzene functionalized porous organic polymers



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